

Updated User Requirements Document (URD)

Echion Health - Version 2.0

Document Control

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1. Introduction

1.1 Purpose

This document defines the user requirements for the Echion Health System, a specialized multi-tenant health application designed to streamline ultrasound scan report creation through:

- AI-powered structured report generation (SonoScribe)
- Personal vault of report templates (SonoVault)
- Hospital-specific customization and branding
- Real-time collaboration and expert consultation (SonoShare)
- Multi-tier subscription model for practices of all sizes

1.2 Scope

The Echion Health System will enable sonographers to create professional ultrasound reports quickly using AI assistance and customizable templates. The system operates as a **multi-tenant platform** where each hospital/clinic has:

- Independent environment with dedicated admin oversight
- Custom branding (letterhead, logo, contact information)
- Team of authorized sonographers and clinicians
- Secure template vault and collaboration space

The platform supports **27+ scan types** including obstetric, abdominal, vascular, cardiac, musculoskeletal, Doppler studies, and specialized examinations. AI-powered report generation using OpenAI 4.0 Mini or Gemini API provides structured findings, impressions, and recommendations.

Key Features:

- **SonoScribe:** AI-assisted report drafting with structured templates
- **SonoVault:** Personal and shared template library
- **SonoShare:** Collaborative case review and consultation
- **Hospital Customization:** Branded reports with custom letterheads
- **Signature Management:** Digital signatures for report authentication
- **Tiered Subscription Plans:** Basic (5 users), Pro (15 users), Ultimate (25 users)

1.3 Intended Audience

This document is intended for:

- Healthcare IT teams
- Software developers and architects
- Quality assurance specialists
- Compliance officers
- Hospital administrators evaluating the system
- Sonographers validating requirements
- Medical facility decision-makers
- Project sponsors and stakeholders

1.4 Business Goals

The Echion Health System aims to:

- **Reduce report generation time by 50-60%** through AI assistance and templates
- Improve diagnostic report quality, consistency, and standardization
- Enable multi-hospital deployment with centralized management

- Provide tiered pricing to accommodate practices of all sizes
 - Facilitate knowledge sharing and peer consultation
 - Ensure HIPAA compliance and data security
 - Support hospital branding and professional presentation
 - Scale to serve 500+ users across multiple healthcare facilities
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2. User Classes and Characteristics

2.1 Hospital Administrator (New Primary Role)

- **Description:** Senior staff member responsible for managing the hospital's Echion Health environment
- **Technical Proficiency:** Intermediate to advanced with healthcare IT systems
- **Frequency of Use:** Weekly for administrative tasks, daily for oversight
- **Key Responsibilities:**
 - User account management (create, authorize, deactivate sonographers)
 - Upload and manage hospital letterhead/logo
 - Authorize signature uploads for clinicians
 - Configure hospital profile (name, address, contact)
 - Monitor system usage and compliance
 - Manage subscription plan and billing
 - Oversee report quality and standardization
 - Grant/revoke access to SonoVault and SonoShare
- **Authority Level:** Full administrative control over organization's environment

2.2 Sonographer (Primary End User)

- **Description:** Certified ultrasound technicians who perform scans and generate diagnostic reports
- **Technical Proficiency:** Basic to intermediate with medical imaging software
- **Frequency of Use:** Daily, multiple times per day (5-20 reports per day)
- **Key Goals:**
 - Quickly generate accurate AI-assisted reports using SonoScribe
 - Maintain personal template library in SonoVault
 - Ensure consistent documentation standards
 - Collaborate on complex cases via SonoShare
 - Produce professionally branded reports with hospital letterhead
- **Clinical Context:** Work in fast-paced environments with limited time between patients

2.3 Radiologist/Consultant (Secondary User)

- **Description:** Medical doctors who review, finalize, and sign off on ultrasound reports

- **Technical Proficiency:** Basic to intermediate
- **Frequency of Use:** Daily for report review and signing
- **Key Goals:**
 - Review sonographer-generated reports
 - Add final impressions and recommendations
 - Apply digital signature to authorize reports
 - Provide expert consultation via SonoShare
- **Signature Authority:** Authorized by admin to sign reports

2.4 System Administrator (IT Support)

- **Description:** IT personnel managing technical infrastructure, security, and system maintenance
 - **Technical Proficiency:** Advanced
 - **Frequency of Use:** As needed for technical administration
 - **Key Goals:** System security, backups, performance monitoring, troubleshooting
-

3. User Requirements

3.1 Functional Requirements

3.1.1 Multi-Tenant Organization Management

UR-001: The system shall support multiple independent hospital/clinic organizations (tenants) with complete data isolation.

- **Priority:** Critical
- **Rationale:** Foundation for multi-tenant SaaS architecture

UR-002: During signup, hospitals shall create an organization account by providing organization name, admin credentials, and contact information.

- **Priority:** Critical
- **Rationale:** Establishes tenant environment

UR-003: Each organization shall have one or more Hospital Administrators with full management authority over their environment.

- **Priority:** Critical
- **Rationale:** Ensures proper governance and oversight

UR-004: The system shall prevent cross-organization data access - users can only access data within their organization.

- **Priority:** Critical
 - **Rationale:** HIPAA compliance and data security
-

3.1.2 Hospital Customization and Branding

UR-005: Hospital Administrators shall be able to upload a custom letterhead/logo (PNG, JPG, SVG formats, max 5MB) for their organization.

- **Priority:** High
- **Rationale:** Professional branding and institutional identity

UR-006: The system shall provide a default Echion Health letterhead template that hospitals can use if they don't upload their own.

- **Priority:** High
- **Rationale:** Ensures all reports have professional headers

UR-007: Hospital Administrators shall be able to configure hospital profile information including:

- Hospital/Clinic name
- Full address
- Contact phone numbers
- Email addresses
- Website URL (optional)
- **Priority:** High
- **Rationale:** Complete contact information on reports

UR-008: The configured letterhead and hospital information shall automatically appear on all generated reports for that organization.

- **Priority:** High
- **Rationale:** Consistent branding across all reports

UR-009: Hospital Administrators shall be able to preview how reports will look with their custom letterhead before finalizing configuration.

- **Priority:** Medium
 - **Rationale:** Quality assurance for branding
-

3.1.3 User Management and Authorization

UR-010: Hospital Administrators shall be able to create user accounts for sonographers and radiologists with the following information:

- Full name
- Username
- Email
- Role (Sonographer, Radiologist, etc.)
- Designation (options: Releasing Radiologist, Consultant Radiologist, Sonographer, Physician Specialist, Cardiologist)
- **Priority:** Critical
- **Rationale:** User access control and role management

UR-011: Users shall authenticate using username and password associated with their organization.

- **Priority:** Critical
- **Rationale:** Security and proper tenant routing

UR-012: The system shall support multi-factor authentication (MFA) as an optional security enhancement.

- **Priority:** High
- **Rationale:** Enhanced security for sensitive medical data

UR-013: Hospital Administrators shall be able to deactivate or reactivate user accounts without deleting historical data.

- **Priority:** High
- **Rationale:** Staff turnover management while preserving audit trails

UR-014: Hospital Administrators shall be able to authorize specific users to upload and manage digital signatures.

- **Priority:** High
- **Rationale:** Control over who can officially sign reports

3.1.4 Digital Signature Management

UR-015: Authorized users shall be able to upload their digital signature (PNG, JPG formats, transparent background preferred, max 2MB).

- **Priority:** High
- **Rationale:** Report authentication and legal compliance

UR-016: The system shall store multiple signatures per user with labels (e.g., "Full Signature", "Initials", "Official Seal").

- **Priority:** Medium

- **Rationale:** Flexibility for different report types

UR-017: When creating or finalizing a report, users shall be able to select which of their uploaded signatures to apply.

- **Priority:** High
- **Rationale:** Appropriate signature for context

UR-018: Signatures shall be rendered on reports along with the user's name and designation beneath the signature.

- **Priority:** High
- **Rationale:** Clear identification of signing authority

UR-019: The system shall maintain an audit log of which signature was used on each report and when.

- **Priority:** Critical
 - **Rationale:** Legal accountability and compliance
-

3.1.5 AI-Powered Report Generation (SonoScribe)

UR-020: The system shall integrate with OpenAI 4.0 Mini API or Google Gemini API for AI-assisted report generation.

- **Priority:** Critical
- **Rationale:** Core differentiating feature for rapid report creation

UR-021: The system shall provide an API endpoint `/api/generate-report` that accepts:

- Raw clinical notes or voice transcript
- Selected scan type (from 27+ predefined types)
- Optional: Measurements and findings data
- **Priority:** Critical
- **Rationale:** AI report generation functionality

UR-022: AI-generated reports shall follow a structured format with clearly defined sections:

- **Findings:** Organ/structure-specific observations
- **Impression:** Summary diagnostic conclusion
- **Recommendations:** (Optional) Suggested follow-up actions
- **Priority:** Critical
- **Rationale:** Clinical documentation standards

UR-023: The system shall implement prompt engineering to ensure AI outputs match medical reporting standards and terminology.

- **Priority:** Critical
- **Rationale:** Clinical accuracy and professionalism

UR-024: Users shall be able to review and edit AI-generated content before finalizing reports.

- **Priority:** Critical
- **Rationale:** Clinical judgment and accuracy verification

UR-025: The system shall provide AI credits as part of subscription tiers:

- Basic: 1,000 credits/month
- Pro: 1,500 credits/month
- Ultimate: 2,000 credits/month
- **Priority:** High
- **Rationale:** Cost management and fair usage

UR-026: The system shall track AI credit usage per organization and alert administrators when approaching limits.

- **Priority:** High
- **Rationale:** Prevent service interruption

3.1.6 Expanded Scan Types and Templates

UR-027: The system shall support the following 27+ scan types with specialized templates:

Abdominal & Pelvic:

1. Abdominal Ultrasound
2. Abdomen and Pelvis (Male)
3. Abdomen and Pelvis (Female)
4. Pelvic Ultrasound (Male)
5. Pelvic Ultrasound (Female)

Obstetric: 6. Obstetric (Early) 7. Obstetric (Late) 8. Obstetric (Twins) 9. Anomaly Scan 10. Biophysical Profile

Gynecological: 11. Transabdominal Pelvic Ultrasound 12. Transvaginal Ultrasound

Cardiac: 13. Echocardiography (Adult Echo) 14. Pediatric Echocardiography

Specialized: 15. Musculoskeletal Ultrasound 16. Neck Ultrasound 17. Thyroid Ultrasound 18. Breast Ultrasound 19. Chest Ultrasound 20. Scrotal (Testicular) Ultrasound 21. Penile Ultrasound 22. Neonatal Head Ultrasound

Vascular/Doppler: 23. Arterial Doppler - Both Lower Limbs 24. Arterial Doppler - Left Lower Limb 25. Arterial Doppler - Right Lower Limb 26. Arterial Doppler - Upper Limbs (Left/Right) 27. Venous Doppler - Both Lower Limbs 28. Venous Doppler - Individual Limbs (Left/Right, Upper/Lower)

General: 29. General Report (customizable free-form template)

- **Priority:** Critical
- **Rationale:** Comprehensive coverage of ultrasound examination types

UR-028: Each scan type template shall include organ/structure-specific sections as defined in the clinical documentation (see document index 3).

- **Priority:** Critical
- **Rationale:** Standardized clinical reporting

UR-029: For Doppler studies, templates shall include velocity measurement tables with findings columns.

- **Priority:** High
- **Rationale:** Quantitative vascular assessment documentation

UR-030: General Report type shall provide an empty customizable space for user-defined scan types not covered by predefined templates.

- **Priority:** High
- **Rationale:** Flexibility for specialized or uncommon examinations

3.1.7 New Report Creation (SonoScribe Interface)

UR-031: Users shall be able to initiate a new report by:

- Selecting scan type from dropdown list
- Choosing to start from scratch OR
- Selecting a template from SonoVault OR
- Using AI-assisted generation (SonoScribe)
- **Priority:** Critical
- **Rationale:** Multiple workflows for different user preferences

UR-032: The new report interface shall include the following mandatory fields:

- **Patient Name**
- **Patient Age**
- **Scan Date** (auto-populated, editable)
- **Scan Type**
- **Priority:** Critical
- **Rationale:** Essential patient identification (to be stripped for template storage)

UR-033: The report form shall present organ/structure-specific input fields based on the selected scan type:

- **Organ/Structure name** (e.g., Liver, Gallbladder, Right Kidney)
- **Findings** (text field for observations)
- **Measurements** (structured numeric fields where applicable)
- **Priority:** High
- **Rationale:** Structured data entry for consistency

UR-034: The system shall provide an "Impression (Summary)" field where users can enter diagnostic conclusions.

- **Priority:** Critical
- **Rationale:** Core clinical documentation requirement

UR-035: The system shall provide a "Recommendations" section with:

- Predefined recommendation options based on scan type (e.g., "MRI recommended for further evaluation", "CT scan of abdomen recommended")
- Free-text field for custom recommendations
- **Priority:** High
- **Rationale:** Clinical follow-up guidance

UR-036: The system shall provide an AI "Suggest Impression" button that generates impression text based on entered findings.

- **Priority:** High
- **Rationale:** AI assistance for diagnostic summary

UR-037: The system shall include "Clinical Differential", "Impression Generator", "Wording Assistant", and "Verified References" helper tools to assist report writing.

- **Priority:** Medium
- **Rationale:** Clinical decision support and education

UR-038: The system shall NOT include voice dictation functionality.

- **Priority:** High
- **Rationale:** Design decision per requirements document

UR-039: The system shall auto-save report drafts every 2 minutes to prevent data loss.

- **Priority:** Critical
 - **Rationale:** Data protection in clinical workflows
-

3.1.8 Report Finalization and Output

UR-040: Before finalizing, users shall be able to preview the complete report with:

- Hospital letterhead
- Patient information
- Findings and measurements
- Impression
- Recommendations
- Clinician signature and designation
- **Priority:** High
- **Rationale:** Quality assurance before report release

UR-041: Users shall select their designation when finalizing a report from the following options:

- Releasing Radiologist
- Consultant Radiologist
- Sonographer
- Physician Specialist
- Cardiologist
- **Priority:** High
- **Rationale:** Proper professional credentialing on reports

UR-042: Users shall select one of their uploaded signatures to apply to the finalized report.

- **Priority:** High
- **Rationale:** Report authentication

UR-043: The system shall automatically validate that all mandatory fields are completed before allowing report finalization.

- **Priority:** High
- **Rationale:** Report completeness and quality

UR-044: Finalized reports shall be exportable in the following formats:

- **PDF** (for printing and EMR integration)
- **DOCX** (for further editing if needed)
- **HL7** (for electronic health record integration)

- **Priority:** Critical
- **Rationale:** Interoperability with healthcare IT systems

UR-045: The system shall render the hospital's custom letterhead at the top of all exported reports.

- **Priority:** High
- **Rationale:** Professional branding compliance

3.1.9 SonoVault - Template Storage & Organization

UR-046: Each user shall have a personal SonoVault for storing and organizing their report templates.

- **Priority:** Critical
- **Rationale:** Personal template library for efficiency

UR-047: Users shall be able to upload existing reports to SonoVault in multiple formats (PDF, DOCX, TXT).

- **Priority:** High
- **Rationale:** Import legacy templates

UR-048: Users shall be able to upload multiple reports simultaneously through bulk upload functionality (CSV, JSON, or multiple files).

- **Priority:** High
- **Rationale:** Efficient initial vault setup

UR-049: When saving a completed report as a template, the system shall automatically:

- Strip patient-identifying information (name, age, dates)
- Flag as "PHI-free template"
- Store in user's SonoVault
- **Priority:** Critical
- **Rationale:** HIPAA compliance for template reuse

UR-050: Users shall be able to organize templates into custom folders/categories by scan type (e.g., "Obstetric - First Trimester", "Abdominal - Liver Pathology").

- **Priority:** High
- **Rationale:** Logical organization for quick access

UR-051: Users shall be able to tag templates with custom labels (complexity level, clinical indication, normal/abnormal).

- **Priority:** High
- **Rationale:** Enhanced searchability

UR-052: The SonoVault interface shall provide:

- Search functionality by keywords, scan type, tags, date range
- Filter by category, favorite status, usage frequency
- Sort options (date, alphabetical, most used)
- Grid or list view with thumbnail previews
- **Priority:** High
- **Rationale:** Efficient template retrieval

UR-053: Users shall be able to mark templates as "Favorite" for quick access.

- **Priority:** High
- **Rationale:** Prioritize frequently used templates

UR-054: The system shall display a "Recently Used" section showing the last 10 accessed templates.

- **Priority:** Medium
- **Rationale:** Workflow efficiency

UR-055: Within SonoVault, users shall be able to navigate to "Upload More Templates" without logging out or losing context.

- **Priority:** Medium
- **Rationale:** Seamless workflow

UR-056: Users shall be able to edit, duplicate, or delete templates in their vault with confirmation prompts.

- **Priority:** High
- **Rationale:** Template lifecycle management

UR-057: The system shall track template version history, allowing users to restore previous versions.

- **Priority:** Medium
- **Rationale:** Safety net for modifications

UR-058: Users shall be able to share selected templates with colleagues within their organization while maintaining ownership.

- **Priority:** High
- **Ratability:** Knowledge sharing within teams

3.1.10 Collaboration Features (SonoShare)

UR-059: Users shall be able to upload in-progress or completed scans to SonoShare for colleague review and expert consultation.

- **Priority:** High
- **Rationale:** Peer consultation on complex cases

UR-060: When sharing a scan, users shall specify:

- Share level (specific colleagues, department-wide, organization-wide)
- Case description and question/concern
- Urgency level (optional)
- **Priority:** High
- **Rationale:** Appropriate collaboration scope

UR-061: Colleagues shall be able to:

- View shared scans and clinical context
- Add comments and feedback
- Suggest diagnostic impressions
- Annotate images (if image integration is available)
- **Priority:** High
- **Rationale:** Meaningful professional dialogue

UR-062: The system shall notify users when colleagues provide feedback on their shared cases.

- **Priority:** Medium
- **Rationale:** Timely awareness of input

UR-063: Users shall be able to mark shared cases as "Resolved" once they have sufficient input.

- **Priority:** Low
- **Rationale:** Case management organization

UR-064: The system shall maintain a complete audit trail of all collaboration activities including who accessed cases and what feedback was provided.

- **Priority:** Critical
 - **Rationale:** Compliance and medicolegal documentation
-

3.1.11 Subscription Tiers and Billing

UR-065: The system shall support three subscription tiers:

Basic Plan (GHC 1,000/month):

- 5 users
- 250 MB storage
- 1,000 AI credits/month
- All core features

Pro Plan (GHC 1,500/month):

- 15 users
- 600 MB storage
- 1,500 AI credits/month
- All core features
- Auto-Grammar Check (NEW)

Ultimate Plan (GHC 2,500/month):

- 25 users
- 1 GB storage
- 2,000 AI credits/month
- All core and admin features
- Auto-Grammar Check
- **Priority:** Critical
- **Rationale:** Scalable business model

UR-066: Organizations shall be able to purchase add-ons:

- More Storage Capacity (from GHC 300/month)
- More AI Credits (from GHC 200/month)
- Lite EMR integration (from GHC 1,000/month)
- **Priority:** Medium
- **Rationale:** Flexible expansion options

UR-067: The system shall enforce user limits based on subscription tier and prevent exceeding licensed user count.

- **Priority:** High
- **Rationale:** License compliance

UR-068: The system shall alert administrators when approaching storage or AI credit limits with options to upgrade.

- **Priority:** High
 - **Rationale:** Prevent service disruption
-

3.2 Non-Functional Requirements

3.2.1 Usability

UR-069: The system shall be accessible from desktop computers, tablets, and mobile devices with responsive design.

- **Priority:** High
- **Rationale:** Diverse clinical environments

UR-070: Creating a new AI-assisted report shall take no more than 3-5 minutes for standard scans for experienced users.

- **Priority:** High
- **Rationale:** Significant time savings over manual reporting

UR-071: New users shall be able to create their first report from a template within 30 minutes of initial training.

- **Priority:** High
- **Rationale:** Low adoption barrier

UR-072: The interface shall be intuitive with clearly labeled sections, minimal clicks to key features, and contextual help.

- **Priority:** High
- **Rationale:** Clinical workflow efficiency

UR-073: The system shall support keyboard shortcuts for frequently performed actions.

- **Priority:** Medium
- **Rationale:** Power user efficiency

UR-074: Medical terminology fields shall support auto-completion.

- **Priority:** Medium

- **Rationale:** Faster data entry
-

3.2.2 Performance

UR-075: AI report generation via API shall complete within 5-10 seconds for typical inputs.

- **Priority:** High
- **Rationale:** Real-time workflow support

UR-076: SonoVault search shall return results within 2 seconds for collections up to 5,000 templates.

- **Priority:** High
- **Rationale:** Efficient template retrieval

UR-077: Report template loading shall occur within 1 second.

- **Priority:** High
- **Rationale:** Seamless workflow

UR-078: The system shall support simultaneous use by:

- Basic tier: 5 concurrent users
- Pro tier: 15 concurrent users
- Ultimate tier: 25 concurrent users without performance degradation.
- **Priority:** High
- **Rationale:** Tier-appropriate scalability

UR-079: File uploads shall support files up to 50MB with progress indicators.

- **Priority:** Medium
 - **Rationale:** Large scan images and reports
-

3.2.3 Security and Privacy

UR-080: The system shall comply with HIPAA regulations for handling protected health information (PHI).

- **Priority:** Critical
- **Rationale:** Legal requirement for healthcare applications

UR-081: All data transmission shall be encrypted using TLS 1.3 or higher.

- **Priority:** Critical
- **Rationale:** Data confidentiality in transit

UR-082: Data at rest shall be encrypted using AES-256 encryption.

- **Priority:** Critical
- **Rationale:** Data protection at rest

UR-083: Multi-tenant data isolation shall be enforced at the database and application layers - no cross-organization data access.

- **Priority:** Critical
- **Rationale:** Tenant security and privacy

UR-084: User authentication shall require strong passwords with complexity requirements (minimum 8 characters, uppercase, lowercase, numbers, special characters).

- **Priority:** Critical
- **Rationale:** Account security

UR-085: The system shall support optional multi-factor authentication (MFA) using authenticator apps.

- **Priority:** High
- **Rationale:** Enhanced security for sensitive environments

UR-086: Role-based access control (RBAC) shall restrict users to appropriate features based on their role (Admin, Sonographer, Radiologist).

- **Priority:** Critical
- **Rationale:** Least privilege principle

UR-087: The system shall automatically detect and flag reports containing PHI when being saved as templates.

- **Priority:** Critical
- **Rationale:** Prevent PHI exposure in shared templates

UR-088: Comprehensive audit logs shall track:

- User authentication (login/logout)
- Report creation, editing, finalization
- Template uploads and access
- Signature applications
- Collaboration activities
- Administrative actions
- **Priority:** Critical

- **Rationale:** HIPAA compliance and security incident investigation

UR-089: User sessions shall timeout after 15 minutes of inactivity.

- **Priority:** High
- **Rationale:** Prevent unauthorized access in shared workstations

UR-090: The system shall support GDPR compliance for international deployments.

- **Priority:** Medium
 - **Rationale:** Global market readiness
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3.2.4 Reliability and Availability

UR-091: The system shall maintain 99.9% uptime during clinical hours (6 AM - 8 PM local time, Monday through Friday).

- **Priority:** High
- **Rationale:** Clinical operations dependency

UR-092: Automatic incremental backups shall occur every 6 hours with full daily backups.

- **Priority:** Critical
- **Rationale:** Data loss prevention

UR-093: Due to auto-save, users shall not lose more than 2 minutes of work in case of system failure.

- **Priority:** High
- **Rationale:** Minimize work loss impact

UR-094: Disaster recovery capabilities shall provide maximum 4-hour Recovery Time Objective (RTO).

- **Priority:** High
 - **Rationale:** Business continuity
-

3.2.5 Data Storage and Scalability

UR-095: Storage limits per tier shall be enforced:

- Basic: 250 MB
- Pro: 600 MB

- Ultimate: 1 GB
- **Priority:** High
- **Rationale:** Subscription tier differentiation

UR-096: The system architecture shall scale to support 500+ active users across multiple organizations.

- **Priority:** High
- **Rationale:** Growth capacity

UR-097: The database architecture shall support efficient multi-tenant queries with tenant-scoped indexes.

- **Priority:** High
- **Rationale:** Performance at scale

UR-098: The system shall support future expansion to include:

- DICOM image integration
- Advanced AI analytics
- EMR/EHR integrations beyond HL7
- Mobile apps (iOS/Android)
- **Priority:** Low
- **Rationale:** Future-proofing platform architecture

4. Updated Database Schema (High-Level)

```
// Multi-tenant organization structure
model Organization {
  id          String  @id @default(uuid())
  name        String
  subscriptionTier SubscriptionTier
  storageLimit Int    // in MB
  aiCreditsLimit Int   // monthly limit
  aiCreditsUsed Int    @default(0)
```

```
// Hospital customization
letterheadUrl String?
hospitalName  String
address       String?
phone         String?
email         String?
website       String?
```

```

    createdAt    DateTime @default(now())
    updatedAt    DateTime @updatedAt

    users        User[]
    vaultTemplates VaultTemplate[]
    reports       Report[]
    sharedScans   SharedScan[]
}

model User {
  id          String @id @default(uuid())
  organizationId String
  username     String @unique
  email        String @unique
  passwordHash String
  fullName     String
  role         UserRole
  designation   Designation?

  // Signature management
  canUploadSignature Boolean @default(false) // Authorized by admin
  signatures           Signature[]

  // MFA
  mfaEnabled Boolean @default(false)
  mfaSecret   String?

  lastLoginAt   DateTime?
  createdAt     DateTime @default(now())
  updatedAt     DateTime @updatedAt

  organization Organization @relation(fields: [organizationId], references: [id], onDelete:
Cascade)
  vaultTemplates VaultTemplate[]
  reports         Report[]
  sharedScans     SharedScan[]
  collaborations Collaboration[]
  auditLogs       AuditLog[]
}

model Signature {
  id          String @id @default(uuid())
  userId      String

```

```

label      String // "Full Signature", "Initials", etc.
imageUrl   String // S3 URL
isDefault  Boolean @default(false)

createdAt  DateTime @default(now())
updatedAt  DateTime @updatedAt

user       User    @relation(fields: [userId], references: [id], onDelete: Cascade)
}

```

```

model Report {
  id          String @id @default(uuid())
  organizationId String
  userId      String
  templateId  String?

  // Patient info (PHI - stored encrypted)
  patientName String @encrypted
  patientAge  Int
  scanDate    DateTime @default(now())

  // Report content
  scanType     ScanType
  findings     Json // Organ/structure findings
  impression   String @db.Text
  recommendations String[] // Selected recommendations

  // Signature
  appliedSignatureId String?
  signatoryName  String?
  designation    Designation?

  // Status
  status      ReportStatus @default(DRAFT)
  aiAssisted  Boolean @default(false)
  aiCreditsUsed Int @default(0)

  lastAutoSaveAt DateTime @default(now())
  completedAt    DateTime?
  exportedAt     DateTime?

  createdAt      DateTime @default(now())
  updatedAt      DateTime @updatedAt
}

```

```
organization    Organization @relation(fields: [organizationId], references: [id])
user            User      @relation(fields: [userId], references: [id])
}
```

```
// Additional enums
```

```
enum SubscriptionTier {
  BASIC
  PRO
  ULTIMATE
}
```

```
enum UserRole {
  HOSPITAL_ADMIN
  SONOGRAPHER
  RADIOLOGIST
  PHYSICIAN
}
```

```
enum Designation {
  RELEASING_RADIOLOGIST
  CONSULTANT_RADIOLOGIST
  SONOGRAPHER
  PHYSICIAN_SPECIALIST
  CARDIOLOGIST
}
```

```
enum ScanType {
  ABDOMINAL
  ABDOMEN_PELVIS_MALE
  ABDOMEN_PELVIS_FEMALE
  PELVIC_MALE
  PELVIC_FEMALE
  OBSTETRIC_EARLY
  OBSTETRIC_LATE
  OBSTETRIC_TWINS
  ANOMALY
  BIOPHYSICAL_PROFILE
  TRANSABDOMINAL_PELVIC
  TRANSVAGINAL
  ECHO_ADULT
  ECHO_PEDIATRIC
  MUSCULOSKELETAL
  NECK
  THYROID
}
```

```
BREAST
CHEST
SCROTAL
PENILE
NEONATAL_HEAD
ARTERIAL_DOPPLER_BOTH_LOWER
ARTERIAL_DOPPLER_LEFT_LOWER
ARTERIAL_DOPPLER_RIGHT_LOWER
ARTERIAL_DOPPLER_LEFT_UPPER
ARTERIAL_DOPPLER_RIGHT_UPPER
VENOUS_DOPPLER_BOTH_LOWER
VENOUS_DOPPLER_LEFT_LOWER
VENOUS_DOPPLER_RIGHT_LOWER
VENOUS_DOPPLER_LEFT_UPPER
VENOUS_DOPPLER_RIGHT_UPPER
GENERAL
}
```

5. Acceptance Criteria (Updated)

5.1 Functional Completeness

- All functional requirements (UR-001 through UR-098) implemented and validated
- Multi-tenant organization signup, configuration, and user management functional
- Hospital letterhead customization and report branding working
- AI report generation via SonoScribe producing clinically accurate structured reports
- All 27+ scan type templates available with proper organ/structure sections
- Digital signature upload, management, and application to reports functional
- SonoVault template storage, search, and bulk upload working
- SonoShare collaboration features enabling case sharing and feedback
- Subscription tier enforcement (user limits, storage, AI credits) operational

5.2 Regulatory and Security Compliance

- HIPAA compliance certified by independent auditor
- Multi-tenant data isolation verified through penetration testing
- Encryption (TLS 1.3, AES-256) implemented and validated
- PHI detection prevents patient data in templates
- Audit logging captures all required activities
- No high-severity security vulnerabilities remaining

5.3 Performance and Reliability

- AI report generation completes in <10 seconds
- SonoVault search returns results in <2 seconds with 5,000 templates
- System supports concurrent users per tier without degradation
- 99.9% uptime achieved during 30-day pilot
- Auto-save prevents data loss in simulated failures

5.4 Usability and User Acceptance

- Hospital admins successfully configure letterhead and manage users
- Sonographers create AI-assisted reports in <5 minutes
- 20+ sonographer UAT achieves 85% satisfaction rating
- 90% of users successfully create reports within 30 minutes of training
- Report outputs with custom letterheads receive positive feedback

5.5 Integration and Interoperability

- PDF, DOCX, and HL7 exports maintain proper formatting
- Custom letterheads render correctly on all exported reports
- Signatures appear professionally on finalized reports

5.6 Clinical Validation

- AI-generated reports reviewed by radiologists for accuracy
 - Report structure and terminology meet clinical standards
 - Collaboration features demonstrate measurable diagnostic confidence improvement
 - No adverse events attributed to system use during pilot
 - Template approach shows measurable improvements in report consistency
-

6. Summary of Key Changes from Version 1.0

1. **Multi-tenant architecture** with organization-level isolation
2. **Hospital Administrator role** with comprehensive management authority
3. **Hospital customization** (letterhead, logo, branding)
4. **Digital signature management** with admin authorization
5. **AI-powered report generation** (SonoScribe) with OpenAI/Gemini integration
6. **Expanded scan types** from ~5 to 27+ specialized templates
7. **Structured organ/structure-based** reporting format
8. **Patient information fields** (name, age) on reports
9. **Clinician designation** selection (5 options)
10. **Recommendations section** with predefined options per scan type

11. **Subscription tier model** (Basic/Pro/Ultimate) with user/storage/AI credit limits
 12. **Add-on services** (storage, credits, EMR integration)
 13. **Auto-grammar check** feature (Pro/Ultimate plans)
 14. **Bulk upload** to SonoVault
 15. **Removal of voice dictation** feature
-

End of Updated URD v2.0

This updated document now reflects the multi-tenant SaaS architecture, hospital customization capabilities, AI-assisted reporting, expanded clinical scope, and tiered subscription model required for Echion Health's evolution into a comprehensive ultrasound reporting platform.